

WolfPack Smart Contract

On-Chain Verification Audit — `src/contracts/WP.h`**PASSED — 16/16 TEST SCRIPTS · 133 ASSERTIONS**

Report date: May 15, 2026 | Target: Qubic Core-Lite testnet (contract index 27)

Methodology: live on-chain transactions via qubic-cli · assertion-based state verification

WolfPack (WP.h) — On-Chain Test Audit

On-chain verification of `src/contracts/WP.h` against a live Qubic Core-Lite testnet node. Every WP procedure, function and system hook was exercised by submitting real transactions and asserting the resulting on-chain state.

- **Contract:** `src/contracts/WP.h` — WolfPack revenue-distribution & staking SC
- **Contract index:** 27
- **Testnet node:** `root@135.181.160.185`, peer port `31841`, HTTP `41841`
- **CLI:** `qubic-cli` (branch `wolfpack-integration`, extended with WP commands)
- **Test harness:** `test-onchain/` — 16 shell scripts, 133 assertions
- **Result:**  all 16 scripts pass

1. Test result summary

#	Script	Asserts	Result
00	<code>00_bootstrap.sh</code>	1	
01	<code>ranks/01_addclan.sh</code>	16	
02	<code>ranks/02_setclanrank.sh</code>	7	
03	<code>ranks/03_removeclan.sh</code>	4	
04	<code>ranks/04_setexclude.sh</code>	16	

#	Script	Asserts	Result
05	<code>ranks/05_clanweightedtotal.sh</code>	10	✓
06	<code>admin/01_setadmin.sh</code>	3	✓
07	<code>admin/02_deposit.sh</code>	4	✓
08	<code>admin/03_distribution_preview.sh</code>	20	✓
09	<code>staking/00_provision.sh</code>	5	✓
10	<code>staking/01_stake.sh</code>	5	✓
11	<code>staking/02_request_unstake.sh</code>	8	✓
12	<code>staking/04_depositstakingrewards.sh</code>	3	✓
13	<code>staking/06_claim_rewards.sh</code>	13	✓
14	<code>staking/03_finalize_unstake.sh</code>	6	✓
15	<code>staking/05_distribution.sh</code>	12	✓

Run the whole suite: `./test-onchain/run.sh`

2. Coverage matrix

Public functions (read-only)

idx	Function	Verified by
1	<code>GetStatus</code>	every script via <code>-wpstatus</code>
2	<code>GetHolderInfo</code>	<code>staking/00_provision</code> , <code>staking/05_distribution</code>
3	<code>GetClanMemberInfo</code>	all <code>ranks/*</code> scripts
4	<code>GetShareholderInfo</code>	<code>-wpshareholderinfo</code>
5	<code>GetStakingInfo</code>	all <code>staking/*</code> scripts
6	<code>GetExcludeAddresses</code>	<code>ranks/04_setexclude</code>
7	<code>GetDistributionPreview</code>	<code>admin/03_distribution_preview</code> (20 asserts)

Public procedures (state-changing)

idx	Procedure	Verified by
1	<code>DepositRevenue</code>	<code>admin/02_deposit</code>
2	<code>AddClanMember</code>	<code>ranks/01_addclan</code>

idx	Procedure	Verified by
3	<code>RemoveClanMember</code>	<code>ranks/03_removeclan</code>
4	<code>SetClanRank</code>	<code>ranks/02_setclanrank</code>
5	<code>SetAdmin</code>	<code>00_bootstrap</code> + <code>admin/01_setadmin</code>
6	<code>SetExcludeAddress</code>	<code>ranks/04_setexclude</code>
7	<code>Stake</code>	<code>staking/01_stake</code>
8	<code>RequestUnstake</code>	<code>staking/02_request_unstake</code>
9	<code>FinalizeUnstake</code>	<code>staking/03_finalize_unstake</code>
10	<code>DepositStakingRewards</code>	<code>staking/04_depositstakingrewards</code>
11	<code>ClaimStakingRewards</code>	<code>staking/06_claim_rewards</code>

System hooks

Hook	What it does	Verified by
<code>INITIALIZE</code>	sets WP-token issuer, zeroes counters	implicit at deploy
<code>BEGIN_EPOCH</code>	snapshots holders + shareholders, distributes staking rewards	<code>staking/00_provision</code> , <code>staking/06_claim_rewards</code>
<code>END_EPOCH</code>	container cleanup	implicit (no crash over many epochs)
<code>END_TICK</code>	daily 11 UTC auto-payout (70/10/10/10)	<code>staking/05_distribution</code>
<code>PRE_ACQUIRE_SHARES</code>	accepts WP-token mgmt transfers from QX	every <code>Stake</code> (mgmt delegation)

Error codes

Verified: `OK(0)` , `ACCESS_DENIED(1)` , `ZERO_AMOUNT(2)` , `NOT_CLAN_MEMBER(4)` , `ALREADY_CLAN_MEMBER(5)` , `INVALID_RANK(6)` , `INSUFFICIENT_STAKE(8)` , `UNSTAKE_PENDING(9)` , `ACQUIRE_FAILED(10)` , `NO_PENDING_REWARDS(12)` , `UNSTAKE_NOT_READY(13)` , `NOT_STAKER(14)` , `INVALID_SLOT(15)` (via state-no-change).

Not exercised: `NOT_HOLDER(3)` , `NO_REWARD(7)` , `TRANSFER_FAILED(11)` — defensive paths that need a deliberately corrupted state to reach.

3. How to verify with qubic-cli

All commands take `-nodeip 127.0.0.1 -nodeport 31841` . On the testnet box:

```
QCLI="/root/qubic/scripts/qubic-cli -nodeip 127.0.0.1 -nodeport 31841"
```

3.1 Contract status

```
$ qubic-cli ... -wpstatus
```

```
===== WolfPack Contract Status =====
Holder Count:          4
Total Tokens Snapshot: 1000000
Shareholder Count:     676
Total Shares Snapshot: 676
Clan Member Count:     0
Clan Weighted Total:   0
Pending Revenue:       0 QU
Reinvestment Fund:     1002472 QU
Total Distributed:     10024690 QU
Total Deposited:       10024690 QU
Last Payout Tick:      48873691
Last Distribution Epoch: 215
Admin:                 WIYVZEQZE0BACHHGYWMLMVXJWVMCUQXJJUKJNSKSTEWJXYTHAJHTNRAENCUIH
```

Sanity check: $\text{Total Deposited} == \text{Total Distributed} + \text{Pending Revenue}$ and $\text{Reinvestment Fund} \approx \text{Total Distributed} \times 10\%$.

3.2 Ranks

```
$ qubic-cli ... -wpclaninfo <ADDRESS>
```

```
===== WolfPack Clan Member Info =====
Address:  TBV0IOBSEJCBUBFUEUNRYLGZCLYAIGMSLLFIMZLHZDAIMNLBBREHFRJARFHJ
Rank:     0
Is Member: NO
```

Admin-only rank management (signed with the admin seed):

```
qubic-cli ... -seed <ADMIN> -wpaddclan    <ADDRESS> <rank 0-5>
qubic-cli ... -seed <ADMIN> -wpsetclanrank <ADDRESS> <rank 0-5>
qubic-cli ... -seed <ADMIN> -wpremoveclan  <ADDRESS>
```

Rank multipliers (permille, from [WP.h](#)): Recruit 1000 · Private 1300 · Sergeant 1800 · Lieutenant 2500 · Colonel 3200 · General 4000. **Clan Weighted Total** in `-wpstatus` is the sum of the multipliers of all current members — e.g. one member of each rank $\Rightarrow$ **13800**.

3.3 Staking

```
$ qubic-cli ... -wpstakinginfo <ADDRESS>

== WolfPack Staking Info ==
Address:      XPKQLZZFGFMWMAXG00WDGDAJFUEDXVQVWJMEE0ESYGIPGLYKINXWWZHGUUVUO
Is Staker:    YES
Staked Amount: 200 WP
Pending Rewards: 0 WP
Unstake Amount: 0 WP
Unstake Epoch: 0
Total Staked (all): 250 WP
Reward Pool:  0 WP
```

Staking lifecycle (signed with the staker's own seed):

```
# 1. delegate WP-share management to the WP contract (index 27) on QX
qubic-cli ... -seed <STAKER> -qxtransferrights WP <WP_ISSUER> 27 <N>
# 2. stake / unstake / finalize
qubic-cli ... -seed <STAKER> -wpstake <N>
qubic-cli ... -seed <STAKER> -wprequestunstake <N>
qubic-cli ... -seed <STAKER> -wpfinalizeunstake # after 2 epochs
qubic-cli ... -seed <STAKER> -wpclaimrewards
```

Verified behaviour:

- **Stake** rejects **0** (**ZERO_AMOUNT**), rejects callers without delegated

WP-management (**ACQUIRE_FAILED**), rejects stakers with a pending unstake (**UNSTAKE_PENDING**).

- **RequestUnstake** rejects non-stakers (**NOT_STAKER**), over-requests

(**INSUFFICIENT_STAKE**), double-requests (**UNSTAKE_PENDING**).

- **FinalizeUnstake** enforces the 2-epoch cooldown (**UNSTAKE_NOT_READY**),

then returns the shares to QX management.

- **BEGIN_EPOCH** distributes the reward pool proportionally — verified split:

pool 600, stakes 50 / 200 $\Rightarrow$ pending rewards 120 / 480.

- **ClaimStakingRewards** clears **Pending Rewards** and releases the reward

shares back to QX management. **Note:** the staker must have at least **staked + pending** WP shares under WP management for the release to succeed — otherwise the claim returns **TRANSFER_FAILED** .

3.4 Distribution preview

```
$ qubic-cli ... -wpdistpreview 1000000

=== WolfPack Distribution Preview for 1000000 QU ===
Holder Share:      700000
Shareholder Share: 100000
```

```
Clan Share:      100000
Reinvest Share:  100000
```

70 / 10 / 10 / 10 split. The remainder of the integer division always lands in `Reinvest Share` (verified with `amount=12345` $\Rightarrow$ 8641 / 1234 / 1234 / 1236).

3.5 Payout (END_TICK auto-distribution)

The daily payout fires inside `END_TICK` when the hour is 11 UTC and at least 1000 ticks have elapsed since the last payout. Deposit revenue, then the next qualifying tick splits it:

```
qubic-cli ... -seed <ANY> -wpdeposit <AMOUNT>      # -> Pending Revenue
# ... END_TICK fires ...
```

Verified with a 10,000,000 QU deposit (`staking/05_distribution.sh`):

Pool	Share	Observed
WP token holders	70%	7,000,000 QU (each 1%-holder +70,000 QU)
SC shareholders (676 IPO)	10%	1,000,000 QU
Clan members (rank-weighted)	10%	999,997 QU
Reinvestment fund	10%	1,000,000 QU

`Total Distributed += amount` , `Pending Revenue  $\rightarrow$  0` , `Last Payout Tick` advances, `Last Distribution Epoch` becomes the current epoch.

3.6 Exclude addresses

```
$ qubic-cli ... -wpexcludeinfo

=== WolfPack Exclude Addresses ===
Exclude Slot 1: AAAA...AAAFXIB      (NULL_ID = unset)
Exclude Slot 2: AAAA...AAAFXIB
```

`-seed <ADMIN> -wpsetexclude <slot 1|2> <ADDRESS>` — admin-only. Slot 0 and slot ≥ 3 are rejected (`INVALID_SLOT`); a non-admin caller is rejected (`ACCESS_DENIED`); both verified by observing the state does not change.

4. Findings during the audit

1. `uint128` $\rightarrow$ `uint64` cast bug (fixed). `(uint64)<uint128_t>` collapses to

`uint128::operator bool()` and yields `1` instead of the quotient. All permille/pro-rata math in `WP.h` (`END_TICK`, `BEGIN_EPOCH`, `GetDistributionPreview`) now reads `uint128.low` explicitly. Guarded by `admin/03_distribution_preview.sh`.

1. **ClaimStakingRewards** **share sourcing**. Rewards are released from the

staker's own WP-managed shares, so a staker must keep `staked + pendingRewards` shares under WP management or the claim fails with `TRANSFER_FAILED`. This is behaviour, not a bug — but worth a CLI warning for end users.

1. **Read-back gaps closed**. `GetExcludeAddresses` (fn 6) and

`clanWeightedTotal` in `GetStatus` were added so exclude slots and the weighted clan total are verifiable on-chain.

5. Test-only modifications (NOT for production)

The following server-side changes make the contract testable in a single session. They must be reverted before any PR / mainnet deployment:

File	Change	Why
<code>contract_def.h</code>	WP deploy epoch <code>210 → 208</code>	deploy WP without waiting 2 epochs
<code>public_settings.h</code>	<code>TESTNET_EPOCH_DURATION 3000 → 500</code>	shorter epochs for cooldown tests
<code>private_settings.h</code>	added test seeds to <code>customSeeds[]</code>	fund operator + issuer + 3 stakers
<code>WP.h</code>	<code>wpToken.issuer</code> → test issuer pubkey	issue the WP asset ourselves on QX
<code>WP.h</code>	<code>END_TICK</code> 11-UTC hour gate commented out	trigger payout any time in tests

`src/contracts/WP.h` changes that **are** intended for the PR: `GetExcludeAddresses` function, `GetDistributionPreview` function, `clanWeightedTotal` field in `GetStatus`, and the `uint128.low` cast fix.